Online Anchor-Robust Forecasting under Economic Regime Shift

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Abstract

Problem. Most online forecasters handle economic regime shifts *reactively*, detecting a change and then recalibrating, and so pay a re-adaptation cost at each break. We study a complementary, proactive strategy for the case in which shifts arrive through an *observable channel* of regime drivers such as geopolitical risk, policy uncertainty or financial volatility. *Method.* We bring anchor regression online: OARF (Online Anchor-Robust Forecasting) is a streaming first-order learner that, with past-only exponential-moving-average moment estimates, drives the component of its residual correlated with a fixed driver channel toward zero; with the penalty set to zero it reduces to online gradient descent. An exploratory add-on, OARF-CD, estimates the driver subspace online from its across-regime nonstationarity. *Theory.* We propose *interventional regret* as an evaluation lens for fixed-channel online anchor robustness and prove the fixed-channel learner is no-regret against a K-switch comparator under linear-SCM and stationarity assumptions ($O(\sqrt{T(1+K)})$ with oracle anchor moments, $O((1+K)T^{2/3})$ online); the learned-channel add-on is *not* covered. *Evidence.* On a controlled structural-causal-model benchmark with a held-out intervention grid, fixed-channel OARF cuts the worst-case interventional error by roughly two-thirds versus reactive and distributionally robust online learners, at the cost of higher in-distribution error; stress tests and additional scenarios show the advantage shrinks and can reverse once the anchor is noisy, misspecified, or not the true regime driver, and the learned channel, though it recovers the subspace on average, is unstable. On sixteen years of daily energy variance-proxy forecasting all methods lie within the Model Confidence Set, so OARF is competitive but not significantly better. The contribution is thus methodological and diagnostic: a streaming anchor-regression update, an interventional-robustness protocol, and evidence on when anchor-based proactive robustness helps or fails.

Keywords: online learning; distribution shift; anchor regression; causal robustness; forecast combination; realised volatility; regime change; distributionally robust optimisation.

JEL: C53, C22, C58, Q47.

1 Introduction

Economic and financial time series rarely keep still. They are punctuated by *regime shifts* — geopolitical shocks, monetary-policy turns, energy crises, and, in carbon markets, discrete regulatory interventions such as the activation of the EU Emissions Trading System Market Stability Reserve

*Corresponding author. Code, data-build scripts and the exact commands that regenerate every number, table and figure are released with this manuscript.

— each of which moves the relationship a forecaster is trying to track. A model that was well calibrated yesterday can be badly miscalibrated tomorrow, and the dominant response in the online-learning and conformal-prediction literatures is to treat this as a detection problem: watch a loss or a coverage statistic, raise an alarm when it drifts, and then recalibrate or re-weight [8, 11, 14]. This reactive posture is intuitive and often effective, but it has two structural weaknesses. It is always one step behind — the alarm can only fire after the loss has already risen — so it pays a re-adaptation cost at every break; and it treats the *symptom* of a shift, a deterioration in fit, as if it were the *cause*.

We take a different view. In many economic settings the cause of a shift is not hidden: it is an *observable channel* of regime drivers whose own changing distribution is precisely what perturbs the forecasting environment. When geopolitical risk spikes or policy uncertainty jumps, the joint distribution of prices, volatilities and their predictors moves with it. If a forecaster's residual can be made *invariant* to that channel — if the part of its error that co-moves with the drivers is held at zero — then the forecaster is protected against shifts in the channel's distribution *before* they happen, without any detection step. This is exactly the logic of anchor regression [15], which trades a little in-sample fit for invariance to interventions on a designated "anchor" variable and thereby minimises a worst-case risk over bounded interventions. Anchor regression, however, is a batch estimator that assumes the analyst already knows which variables constitute the anchor. Neither assumption fits a streaming economic deployment, where data arrive one observation at a time and where the right channel — which of the many candidate drivers actually carries the regime — is itself uncertain.

This paper closes both gaps. We develop OARF, a fully streaming anchor-robust forecaster whose first-order update augments the ordinary prediction gradient with an anchor-robustness term assembled from exponential-moving-average cross-moments; with the robustness weight set to zero it degenerates exactly to online gradient descent, so robustness is a single, interpretable dial. On top of it we build a second layer that *learns the channel from the stream*, identifying the subspace of the observable context along which the relationship to the regime is most unstable, and feeding that learned anchor back into the robust update on a slower timescale. Finally, because the usual notions of regret do not capture what such a method is for, we define and analyse *interventional regret*: regret measured against the best regime-conditional predictor under the worst-case bounded future intervention.

Contributions.

1. **An online implementation of anchor regression** (Section 3). OARF realises the anchor-regression objective as a streaming first-order update whose robustness term is built from EMA cross-moments of centred variables. This is a faithful online counterpart of an existing batch estimator rather than a new objective; setting $\xi = 0$ recovers online gradient descent.

2. **A heuristic online channel-discovery layer** (Section 3). A two-timescale rule estimates which directions of the observable context behave as the anchor by tracking their across-regime nonstationarity. This layer is heuristic, that it identifies a nonstationary subspace rather than a *certified* causal channel and that it is *not* covered by our theorem.

3. **Interventional regret for the fixed-channel learner** (Section 3). We define interventional regret and give a complete proof that the *fixed*-channel update is no-regret against a K-switch comparator: $O(\sqrt{T(1+K)})$ with oracle anchor moments and $O((1+K)T^{2/3})$ with online moment estimates. The learned-channel variant is not analysed.

4. **An empirical study with stress tests and honest limits** (Sections 4–5). On a controlled SCM with a held-out intervention grid we measure interventional robustness against ground truth and, through stress tests, locate the point at which a noisy or misspecified anchor makes the method *worse* than no anchor; we then study forecast combination on sixteen years of public energy data against ten baselines with Diebold–Mariano [5] and Model Confidence Set [9] significance.

The empirical picture is mixed by design. On data whose shifts travel through a clean, observed channel, proactive robustness trades a higher in-distribution error for a large reduction in worst-case interventional error; on real energy series, where that channel is weak, the method is competitive but not significantly better than the strongest baselines. We report both outcomes, and stress tests (Section 5) map where the approach fails.

2 Related work

Our method sits at the meeting point of three literatures, and it is easiest to position by what it borrows from and where it departs from each.

Causal robustness and invariance. The intellectual root is the observation that a predictor which is *invariant* across environments is also robust to interventions that move between them. Invariant causal prediction [13] makes this precise for the identification of causal parents, and anchor regression [15] turns it into an estimator: by penalising the projection of the residual onto an anchor variable it interpolates between ordinary least squares and an instrumental-variables solution, and the penalised problem is provably equivalent to minimising worst-case risk over a bounded set of interventions on that anchor. DRIG [16] generalises the principle to distributional robustness through invariant gradients. What all of these share — and what we must overcome for a streaming deployment — is that they are batch procedures that take the environment or anchor variable as given. OARF keeps the worst-case-intervention semantics of anchor regression but makes the estimator online and, crucially, learns the anchor channel from data.

Reactive online adaptation. A second, very active line treats distribution shift as something to detect and then accommodate. Adaptive conformal inference [8] adjusts a miscoverage level online to maintain coverage under drift. The 2025 state of the art sharpens the detect-then-adapt loop considerably: M-FISHER [11] maintains an exponential test martingale over non-conformity scores and, when the martingale crosses a Ville threshold, takes a Fisher-preconditioned natural-gradient correction; WATCH/WCTM [14] monitors a *weighted* conformal test martingale that can tolerate benign covariate shift while still flagging harmful concept shift, and adapts the prediction interval accordingly. These methods are the natural incumbents against which a proactive method must be judged — and the comparison is precisely the point, since OARF aims to make the trigger unnecessary by removing the channel's influence before it can raise the loss.

Distributionally robust forecasting (2026). The closest competitors are recent methods that, like OARF, optimise an explicitly worst-case objective. Wang [18] propose online randomised distributionally robust forecast combination for dependent data, drawing combination weights from a distribution over the simplex and updating it by stochastic mirror descent on a Wasserstein ambiguity objective. Liu and Cheng [12] develop adaptive robust combination with variance- and Expected-Shortfall-scaled exponential weights for yield-curve forecasting, and Chen et al. [3] analyse

Wasserstein distributionally robust *online* learning as a primal–dual game with regret guarantees; a parallel strand makes conformal decisions cost-sensitive [19]. Several of these 2025–2026 works are recent preprints; we treat them as the contemporary literature and competitive baselines rather than as settled, peer-reviewed results. The key distinction from OARF is geometric. These methods hedge over an *isotropic* ambiguity ball centred at the empirical distribution: they spread robustness equally in all directions, including directions along which the world never actually moves. OARF instead hedges over the *anisotropic, anchor-shaped ellipsoid* $\mathbb{E}[aa^\top]$, spending its robustness budget only along the anchor directions where shifts genuinely travel. Our experiments are designed to isolate exactly this difference, and they show it to be the source of OARF's advantage.

Adjacent online and subspace methods. Three nearby literatures inform our design. *Online invariant / domain-adaptation* methods seek predictors that hold across changing environments in a streaming setting; OARF can be seen as an online, single-anchor instance of this idea specialised to mean-shift interventions, with a closed-form robustness term rather than an explicit environment partition. *Nonstationary-subspace* and change-adaptive analysis (subspace tracking, stationary-subspace analysis, and the path-length-adaptive online learning of 20) is exactly what our discovery layer performs — it is an unsupervised nonstationary-subspace estimator — which is why we are careful not to call its output causal. The distinction from all three is that OARF couples the discovered nonstationary subspace back into an anchor-regression objective with an interventional, rather than purely predictive, target.

Online convex optimisation. Methodologically, the regret analysis rests on dynamic-regret results for online gradient descent [10, 21], the path-length-adaptive refinement of Zhang et al. [20], and AdaGrad-style preconditioning [6]. The economic measurements draw on the daily Geopolitical Risk index [2], the Economic Policy Uncertainty index [1], realised-volatility modelling in the heterogeneous-autoregressive tradition [4], and the volatility-timing logic of Fleming et al. [7].

3 The OARF method and its guarantee

Setup and a leak-free protocol. We work in discrete time $t = 1, \dots, T$. At each step the forecaster observes predictors $x_t \in \mathbb{R}^d$ — which in a pure forecasting problem are explanatory variables and in a combination problem are the forecasts of several base models — together with a vector of candidate regime drivers $z_t \in \mathbb{R}^p$, and it must predict a scalar target y_t. A linear predictor with feature map $\phi(x) = [1; x] \in \mathbb{R}^{d+1}$ and weights w emits $\hat{y}_t = w^\top \phi(x_t)$, incurs the squared loss $\ell_t(w) = (y_t - \hat{y}_t)^2$, and writes its residual $r_t = y_t - \hat{y}_t$. The *anchor* is a (possibly learned) linear read-out of the context, $a_t = B^\top z_t \in \mathbb{R}^q$, with channel matrix $B \in \mathbb{R}^{p\times q}$. The entire pipeline is leak-free: at step t we first predict from x_t, then reveal y_t and update, and every standardisation uses past-only statistics; when we later evaluate a frozen learner on held-out interventions we replay exactly that frozen, past-only standardiser, so no future information can ever enter a forecast.

The causal-robustness objective. Following anchor regression [15], OARF trades raw predictability against invariance of the residual to the anchor by minimising

$$b_{\mathrm{AR}}(w;\xi) = \underbrace{\mathbb{E}\big[(y - w^\top \phi(x))^2\big]}_{\text{predictive loss}} + \xi\, \underbrace{\mathbb{E}[ar]^\top \big(\mathbb{E}[aa^\top]\big)^{-1} \mathbb{E}[ar]}_{P(w)=\|\operatorname{proj}_a r\|^2}, \qquad \xi \geq 0. \tag{1}$$

The penalty $P(w)$ is the squared length of the projection of the residual onto the anchor; driving it down forces $\mathbb{E}[ar] \to 0$, so the predictor stops exploiting any component of the signal that co-moves with the regime drivers. At $\xi = 0$ the objective is ordinary least squares, and as ξ grows the solution moves toward the invariant predictor. The reason this is the right object is not merely intuitive:

Proposition 1 (Interventional-robustness equivalence, 15, Thm. 1)**.** *Assume a linear structural causal model in which (a) the anchor A is a* source *(exogenous) node; (b) the remaining variables, including the target, follow linear structural assignments $V = \mathbf{B}V + \varepsilon + \mathbf{M}A$ with $A \perp \varepsilon$ and $I - \mathbf{B}$ invertible; (c) the considered interventions are mean-shift interventions* $\mathrm{do}(a := \nu)$ *that replace A by a constant; and (d) the loss is squared error. Then for every $\xi \geq 0$ there is a radius $\rho(\xi)$, given in closed form by the cited theorem and strictly increasing in ξ (with $\rho(0) = 0$, the OLS support, and $\rho(\xi) \to \infty$ as $\xi \to \infty$, the instrumental-variables limit), such that*

$$\min_w b_{\mathrm{AR}}(w; \xi) \;=\; \min_w \sup_{\nu:\, \|\nu\| \leq \rho(\xi)} \mathbb{E}_{\mathrm{do}(a:=\nu)}\big[(y - w^\top \phi(x))^2\big]. \tag{2}$$

The equivalence is thus not automatic for arbitrary nonstationarity: it requires the exogenous-anchor SCM (a)–(c) above, and the worst case is over *mean-shift* interventions specifically. Under these conditions the set hedged over is an ellipsoid shaped by the anchor covariance $\mathbb{E}[aa^\top]$ — not the isotropic ball of the distributionally robust methods of Section 2 — which is the geometric distinction our evaluation sets out to test. Our theory inherits exactly these assumptions (Assumption 2); where they fail, the stress tests of Section 5 quantify the degradation.

A two-signal example. The intuition is simplest with two predictors: a stable *parent* x_{p} that genuinely drives y, and a *child* x_{c} that is generated from y but is also pushed around by the anchor, $x_{\mathrm{c}} = c\,y + \gamma a + \text{noise}$. In a calm period both are informative, and OLS happily loads on x_{c} because it correlates strongly with y — so in-distribution it wins. But the child's usefulness is borrowed from the current anchor level: when the regime pushes a to a new value, the γa term shifts and the child's apparent relationship to y breaks, so the OLS forecast degrades. Anchor regression sees that the residual of an x_{c}-heavy predictor co-moves with a and penalises it, shifting weight onto the parent; it gives up a little calm-period accuracy to avoid the break. OARF performs this trade online, and the synthetic benchmark below is exactly this mechanism scaled up to several parents and children.

A streaming anchor-gradient update. The gradient of (1) in the coefficient block separates into a prediction term and a robustness term,

$$\nabla_w b_{\mathrm{AR}} = -2\,\mathbb{E}[\phi r] \;-\; 2\xi\,\mathbb{E}[\phi a^\top]\big(\mathbb{E}[aa^\top]\big)^{-1}\mathbb{E}[ar], \tag{3}$$

and the move from batch to streaming is to replace each population moment by an exponential moving average (decay β) of its sample analogue, computed on *centred* variables. The role of centring is more limited than a slogan might suggest. Subtracting the running means removes the *across-regime mean component* of the anchor; under the SCM of Proposition 1, where a regime is a mean-shift on A, this is the component that varies between regimes, so what remains is the within-regime second-moment structure the robustness term (2) uses. We do *not* claim centring removes the intervention itself — the intervention's effect persists in X and Y — only that it prevents the slowly-varying regime mean from contaminating the covariance estimate; the channel-discovery layer below deliberately uses the across-regime mean variation that centring discards. Writing $\tilde{\cdot}$ for the EMA-centred quantities, and collecting the non-intercept coefficients as w_{-0}, we maintain running estimates m_t^{Xr}, m_t^{XA}, M_t^{AA}, m_t^{Ar} of $\mathbb{E}[\tilde{\phi}\,\tilde{r}]$, $\mathbb{E}[\tilde{\phi}\,\tilde{a}^\top]$, $\mathbb{E}[\tilde{a}\tilde{a}^\top]$, $\mathbb{E}[\tilde{a}\tilde{r}]$ (here $\tilde{\phi}$ stacks the intercept and centred $\tilde{x}$, and the intercept row of m_t^{XA} is held at zero so the penalty never acts on the intercept), and update

$$\boxed{w_{t+1} = w_t - \eta\, g_t, \qquad g_t = \underbrace{-\, r_t\, \phi(x_t)}_{\text{prediction}} \underbrace{-\, 2\xi\, [\, 0;\ m_t^{XA}(M_t^{AA} + \epsilon I)^{-1} m_t^{Ar}\,]}_{\text{anchor robustness}} + \lambda_2[0; w_{t,-0}].} \tag{4}$$

The two factors of -2 from (3) are absorbed into the step size η (prediction) and kept explicit (robustness) only to make the relative weight ξ read directly; signs follow gradient *descent*, so the ridge term $+\lambda_2[0; w_{-0}]$ enters g_t with a positive sign and the update $w \leftarrow w - \eta g_t$ therefore *shrinks* the non-intercept weights, as intended. The ridge ϵ stabilises the anchor Gram M_t^{AA} when it is ill-conditioned (we use $\epsilon = 10^{-3}$); per step the update costs $O(d + pq + q^3)$, dominated by the $q \times q$ solve, which is negligible for the small q of interest. In practice we take the step with AdaGrad per-coordinate scaling [6]: the *gradient* g_t is exactly (4) and only its step length is preconditioned, so the online-convex-optimisation regret machinery is unchanged while one global rate adapts to features of very different scale. With $\xi = 0$, and regardless of the AdaGrad and standardisation machinery (which act identically on all methods), g_t reduces to the plain squared-loss gradient and the update is online gradient descent. The decay β controls how quickly the moment estimates forget.

Discovering the channel online (a heuristic). The development so far assumes the analyst supplies B. We now describe a heuristic that estimates B from the stream; we stress at the outset that it is *not* covered by the theorem of the next subsection and that it performs *unsupervised nonstationary-subspace estimation*, not causal discovery. The motivation is that, under the SCM of Proposition 1, a regime acts on the context by shifting the anchor's location, so the anchor directions are nonstationary in mean across regimes while stationary nuisance directions are not. We therefore track the across-regime dispersion of the context's conditional mean and ascend it on a slow timescale $\eta_B \ll \eta$,

$$\max_{\|B\|_F \le 1} \ \mathcal{D}(B) = \operatorname*{Var}_{\text{regimes}} \big(\mathbb{E}[B^\top z \mid \text{regime}]\big) - \gamma_c \, \big\|\mathbb{E}[B^\top z]\big\|^2. \tag{5}$$

This objective selects *high-dispersion nonstationary* directions; it does not, by itself, certify that they are exogenous or causal. The grand-mean penalty (second term) discourages loading on the persistent component but does not guarantee exogeneity, and a direction that is merely volatile or trending can in principle be selected — a failure mode we probe directly in the stress tests of Section 5. The method is therefore best read as recovering the subspace through which regimes *appear* to act in the observable context, which coincides with the causal channel only under the SCM assumptions. Streaming (5) needs a proxy for "regime": we use a forgetting partition into length-L blocks, long enough that a stationary driver's short-horizon autocorrelation averages out within a block (so L should exceed the dominant driver autocorrelation time) yet short enough that genuine regime changes fall across block boundaries; L is a hyperparameter selected on the warm-up window, and we report sensitivity to it in Section 5. Letting S be the across-block scatter of the block-mean context and u its grand mean, the ascent direction is $\nabla_B \mathcal{D} = 2(S - \gamma_c u u^\top)B$; at each block boundary B is updated *after* the robust gradient step of that step and re-orthonormalised onto the Frobenius ball, and the new B is used from the next step onward. A lightweight non-robust reference predictor is carried alongside to keep discovery oriented toward predictively relevant directions. Algorithm 1 states the complete loop, with the discovery layer as an optional inner block.

Algorithm 1 OARF with online channel discovery (one streaming pass)

input: penalty ξ, rates η, η_B, decay β, ridge ϵ, λ_2, block L, channel B (or random init if learning)
$w \leftarrow 0$; EMA means and cross-moments $\leftarrow 0$; block buffer $\leftarrow 0$
for $t = 1, \ldots, T$ **do**
  observe x_t, z_t; standardise (past-only, winsorised): x_s, z_s; $a \leftarrow B^\top z_s$
  $\hat{y}_t \leftarrow w^\top [1; x_s]$; **emit** $\hat{y}_t$; reveal y_t; $r \leftarrow y_t - \hat{y}_t$
  update EMA means; centre $\tilde{x}, \tilde{a}, \tilde{r}$; update $m^{Xr}, m^{XA}, M^{AA}, m^{Ar}$
  intercept gradient $g_0 \leftarrow -r$
  coef. gradient $g_{-0} \leftarrow -r\, x_s - 2\xi\, m^{XA}(M^{AA} + \epsilon I)^{-1} m^{Ar} + \lambda_2\, w_{-0}$
  $w \leftarrow \Pi_{\mathcal{W}}\big(w - \textsc{AdaGradStep}(g)\big)$ ▷ $g = [g_0; g_{-0}]$; ridge sign shrinks w_{-0}
  if learning the channel **then**
    accumulate z_s into the block buffer; **every** L steps update across-block scatter S, mean u, and set $B \leftarrow \mathrm{orth}\big(B + \eta_B\, 2(S - \gamma_c u u^\top)B\big)$
  end if
  absorb (x_t, z_t) into the standardiser statistics
end for

Dimensions: $w \in \mathbb{R}^{d+1}$ (w_0 intercept, $w_{-0} \in \mathbb{R}^d$); $m^{Xr}, x_s \in \mathbb{R}^d$; $m^{XA} \in \mathbb{R}^{d\times q}$ (intercept row excluded); $M^{AA} \in \mathbb{R}^{q\times q}$; $m^{Ar} \in \mathbb{R}^q$; channel $B \in \mathbb{R}^{p\times q}$. $\Pi_{\mathcal{W}}$ projects onto the radius-D ball.

Interventional regret. To state what OARF guarantees we need a regret notion that rewards robustness to *future* interventions, not merely good tracking of the past. For a radius $\rho \geq 0$ define

$$\mathrm{IntReg}_T(\rho) = \sum_{t=1}^{T} \ell_t(w_t) - \min_{u \in \mathcal{W}} \sum_{t=1}^{T} \sup_{\|\nu_t\| \leq \rho} \mathbb{E}_{\mathrm{do}(a:=\nu_t)}\big[(y_t - u_t^\top \phi(x_t))^2\big] : \tag{6}$$

the learner's realised loss, measured against the best comparator in a class $\mathcal{W}$ that is itself stress-tested by the worst bounded intervention at every step.

Why this regret notion, and what it does not imply. Ordinary dynamic regret asks the learner to track the best predictor *on the realised sequence*, which rewards fitting whatever just happened — exactly the reactive behaviour we wish to move beyond — and says nothing about robustness to a shift that has not yet occurred. Consider two regimes that differ only in the anchor's location: a predictor that exploits an anchor-correlated signal can have low dynamic regret (it tracks each realised regime well) yet incur large loss the moment the anchor is pushed to a new value. Interventional regret instead holds the comparator to a higher standard — it must do well under the *worst* bounded intervention — so a learner can only compete by being robust across the whole intervention class, not just on the realised path. The asymmetry (the comparator is stress-tested, the learner is scored on realised loss) is deliberate and *favours the comparator*, making the bound harder to attain, not easier; the realised stream is itself one admissible intervention (Assumption 2), so the learner is implicitly stressed by the shifts it actually meets. The guarantee does *not* imply: superiority under arbitrary adversarial shifts (the comparator class is regime-conditional, not adversarial); any guarantee for the learned channel; or low realised MSE when the anchor is wrong — the stress tests and scenarios show that last case explicitly.

We make four assumptions, each used in a specific place.

Assumption 1. *(Boundedness.) The standardised, winsorised features and anchor are bounded, so the per-step gradients of the surrogate below are bounded in norm by a constant G, and the comparator class $\mathcal{W}$ lies in a Euclidean ball of radius D.*

Assumption 2. *(Linear-SCM anchor model.) The data follow a linear structural causal model in which the anchor enters as in Section 3, so the equivalence of Proposition 1 holds within each regime, and the radius is adequate, $\rho(\xi) \geq \max_t \|\nu_t^\star\|$, where $\nu_t^\star$ is the realised regime's anchor displacement (so each realised regime is one admissible intervention in the ball).*

Assumption 3. *(Oblivious regime schedule and comparator.) The regime schedule is exogenous and oblivious to the learner; the per-step worst-case risk $\mathcal{R}_t$ defined below is therefore a deterministic function of w, and $\mathcal{W}$ is the class of regime-conditional comparators that switch at most K times (path length $P_T \leq 2DK$).*

Assumption 4. *(Within-regime stationarity.) Within each regime the joint distribution of (ϕ, a, y) is stationary with mixing fast relative to the EMA window, so the EMA moment estimates obey the concentration of Lemma 3.*

We view Algorithm 1 as online gradient descent fed *inexact* gradients on the convex per-step worst-case risks $\mathcal{R}_t(w) := \sup_{\|\nu\|\leq\rho(\xi)} \mathbb{E}_{\mathrm{do}(a:=\nu)}[(y_t - w^\top\phi(x_t))^2]$ of (6). The full argument, organised as three lemmas, is deferred to Appendix A; we state the result and sketch it here.

Theorem 1 (Fixed-anchor OARF interventional regret)**.** *Under Assumptions 1–4, Algorithm 1 with a fixed channel and the Ader step wrapper attains* sublinear *expected interventional regret against any K-switch comparator, with no prior knowledge of K:*

(i) with oracle anchor second moments, $\mathbb{E}\big[\mathrm{IntReg}_T(\rho(\xi))\big] = O\big(\sqrt{T(1+K)}\big)$*;*

(ii) with the EMA moment estimates of Algorithm 1 and decay $1-\beta_T = T^{-2/3}$*,* $\mathbb{E}\big[\mathrm{IntReg}_T(\rho(\xi))\big] = O\big((1+K)\,T^{2/3}\big)$.

Proof sketch (full proof in Appendix A). Three steps. (1) $\mathcal{R}_t$ is convex and, by Proposition 1, equals the anchor-penalised risk within a regime; the update direction is $\nabla\mathcal{R}_t(w_t) + \zeta_t$, where ζ_t splits into a zero-mean martingale part M_t (one-sample gradient noise) and a deterministic part b_t (EMA moment-estimation error). (2) The path-length-adaptive Ader wrapper [20] gives dynamic regret $O(\sqrt{T(1+K)})$ against a K-switch comparator without knowing K, plus the inexactness term $\sum_t\langle\zeta_t, w_t - u_t\rangle$. (3) Against the deterministic (oblivious-schedule) comparator, $\mathbb{E}\sum_t\langle M_t, w_t - u_t\rangle = 0$, while $D\sum_t \mathbb{E}\|b_t\| \leq D(c_1 T\sqrt{1-\beta} + c_2 K/(1-\beta))$ is zero with oracle moments and $O((1{+}K)T^{2/3})$ at $1{-}\beta_T = T^{-2/3}$. Bounding realised loss by the worst-case risk, $\mathbb{E}_t[\ell_t(w_t)] \leq \mathcal{R}_t(w_t)$ (Assumption 2), gives the two rates; both are $o(T)$. □

Remark 1 (Reading the two rates). The gap between (i) and (ii) is the price of *learning the robustness geometry online without changepoint detection.* A detection-and-reset estimator could drive the moment error down at the $O(1/\sqrt{n})$ within-regime rate and recover (i), but only by reintroducing exactly the triggers OARF is designed to avoid; the $T^{2/3}$ rate is what a purely proactive, exponentially-forgetting estimator pays, and it remains sublinear. The deployed implementation uses a single AdaGrad instance rather than the Ader wrapper, which removes only the K-adaptivity and is what one wants in practice.

Remark 2 (What the guarantee does and does not claim). Two qualifications matter. First, risk-adjusted no-regret is attainable only under stochastic, stationary or ergodic data, not against a fully adversarial environment [17]; this is why $\mathcal{W}$ is regime-conditional rather than adversarial and why observable regime drivers must enter the statement — they define the regimes the comparator may condition on. Second, anchor-based methods deliver *interventional robustness, not causal identification*: the estimand is the diluted-causal parameter, and we do not claim to recover a structural coefficient.

4 Experimental design

We evaluate OARF in two complementary arenas. A controlled structural-causal-model benchmark lets us measure interventional robustness against ground truth, because we can intervene on the anchor by construction; a sixteen-year real-data study then asks whether the mechanism survives contact with genuine, messy economic series, where no ground-truth intervention exists and the regime structure is milder. The two together are designed so that the synthetic study isolates *why* the method should help and the real study tests *whether* it does.

A controlled anchor-SCM with a measurable intervention gap. The synthetic generator (Table 1) is built so that the difference between ordinary prediction and interventional robustness is not a matter of taste but a quantity we can read off. An exogenous anchor A, whose mean shifts across eight regimes — those shifts *are* the in-sample interventions — and a hidden confounder H drive a block of causal *parents* X_{par} of the target, which is generated as $Y = X_{\text{par}}^{\top} b_{\text{par}} + H^{\top} d_H + \varepsilon_Y$. A second block of *children* is then generated downstream of the target, $X_{\text{ch}} = Y c_{\text{ch}} + A W_{Ac} + H W_{Hc} + \varepsilon_{\text{ch}}$, so that the children are anti-causal predictors that also load on the anchor. The design has a deliberate trap. In-sample the children are extremely informative about Y — they are generated from it — so ordinary least squares and online gradient descent lean on them heavily; but a child's relationship to Y is contaminated by the anchor term AW_{Ac}, and under an intervention $\text{do}(A := \nu)$ that term is set externally, so the child turns from informative to misleading and injects an error that grows with $\|\nu\|$. The only predictor that survives the intervention is the one that uses the parents alone, and that is precisely the gap anchor robustness is meant to close. To give the discovery layer something to recover, the observable context z is a random rotation of the anchor together with $p - q$ stationary AR(1) decoy drivers, so that there is a ground-truth channel B_{true} with $ZB_{\text{true}} = A$. We hold out a frozen grid of interventions — ten magnitudes, six directions, four hundred draws each — spanning anchor values *beyond* the training range, and it is on this grid that every interventional number in Section 5 is computed.

Table 1: Configuration of the synthetic anchor-SCM (defaults; results average over 8 random seeds).

Symbol	Meaning	Value
T	stream length	8000
$d_{\text{par}}, d_{\text{ch}}$	causal parents / anti-causal children	$3, 3$
p	observable context dimension	8
q	true anchor (channel) dimension	2
h	hidden-confounder dimension	2
#regimes	anchor-mean regimes ($\Rightarrow$ 7 changepoints)	8
σ_{shift}	across-regime anchor-mean scale	1.5
σ_A	within-regime anchor sd	0.6
c_{ch} scale	strength of the target$\to$child edge	1.3
W_{Ac} scale	anchor$\to$child contamination	1.1
confounding	strength of the $H \to (X, Y)$ path	0.9
σ_X, σ_Y	predictor / target noise	$0.4, 0.5$

The real panel and the forecasting task. For the real study we assemble a public daily panel spanning 2010–2026 (Table 2), aligned to a business-day calendar with capped forward-fill. The targets are Brent crude and Henry Hub natural gas, and the prediction problem is one-step-ahead *daily realised-variance proxy.* We use the term carefully: with only daily closing prices we cannot form true intraday realised variance, so the target is the log of a five-day rolling mean of squared daily log-returns — a coarse, daily-frequency variance proxy, not an intraday RV estimator. We forecast variance rather than returns deliberately: daily returns are close to unforecastable, so a return study would mostly measure noise, whereas this variance proxy is both predictable (volatility clustering) and regime-dependent, which is where regime-robustness can matter. The macro drivers — the CBOE VIX, the dollar, the ten-year yield, and the daily Geopolitical Risk and Economic Policy Uncertainty indices — serve as the observable context. We use VIX strictly as a *cross-asset uncertainty driver*, not as a commodity implied-volatility measure (it is equity-index implied volatility); a commodity-specific implied-vol series would be preferable but is not freely available at daily frequency over this span. The levels of the three uncertainty indices form the hand-chosen anchor for the fixed-channel variant, while OARF-CD is left to find its own. Regimes for the regime-aware metrics are defined, without look-ahead, as terciles of a past-only volatility-stress index, giving calm, normal and turbulent states and the transitions between them.

Table 2: The public real-data panel. All series are free and citation-requested and are fetched by the released download script.

Series	Source / ticker	Span (daily)	Role
Brent crude	FRED `DCOILBRENTEU`	2010–2026	target (log-RV)
Henry Hub gas	FRED `DHHNGSP`	2010–2026	target (log-RV)
VIX	FRED `VIXCLS`	2010–2026	driver (level + change)
EUR/USD	FRED `DEXUSEU`	2010–2026	driver
10y Treasury	FRED `DGS10`	2010–2026	driver
Geopolitical Risk	Caldara–Iacoviello [2]	2010–2026	driver / anchor
Policy Uncertainty	Baker–Bloom–Davis [1]	2010–2026	driver / anchor

The forecast-combination pool. In the real study every method combines the same pool of one-step-ahead volatility forecasts, and the pool is chosen to be the standard one rather than anything bespoke. It comprises the heterogeneous-autoregressive cascade of daily, weekly and monthly lagged variance [4]; an exponentially weighted RiskMetrics estimator; a GARCH$(1,1)$ conditional variance; (log) VIX as a cross-asset uncertainty signal; and a regularised regression on these together with the macro drivers. These are not interchangeable: HAR encodes long-memory persistence, the EWMA and GARCH terms capture short-run clustering with different decay structures, and the VIX signal injects cross-asset information the autoregressive models miss. No single one dominates across regimes, which is precisely why combining them is worthwhile and why this is the natural arena in which the 2026 distributionally robust combiners were proposed. Every member of the pool is strictly causal: each forecast for day t is computed from information available through $t-1$.

Baselines. We compare against eleven methods, grouped not by superficial type but by the question each is meant to answer about OARF (Table 3). The *basic baselines* — online gradient descent, which is literally OARF with $\xi = 0$; a rolling-window ridge regression; and adaptive conformal inference [8] — pin down what accuracy looks like with no robustness mechanism and what a strong batch refit can do, and so calibrate both ends of the scale. The *2025 reactive* methods, M-FISHER [11] and WATCH/WCTM [14], embody the detect-then-adapt paradigm that OARF is conceived to replace; placing them side by side tests directly whether removing a channel's influence in advance beats reacting to it after the loss rises. The *2026 distributionally robust* methods — randomised Wasserstein combination [18], the variance- and ES-scaled exponential-weight combiners [12], Wasserstein online learning [3] and cost-aware conformal inference [19] — are the closest comparison, because they too optimise a worst-case objective; what separates them from OARF is only the *shape* of the set they hedge over, an agnostic ball rather than an anchor-shaped ellipsoid, so the comparison isolates the value of that geometry. Finally, the *batch causal-robust* estimators, anchor regression [15] and DRIG [16], use the same invariance principle as OARF but offline and with the channel handed to them; they therefore serve as an oracle upper bound on what the on-line, channel-learning version can hope to match. Every method is run under the identical leak-free protocol, with AdaGrad-stabilised steps where applicable.

Table 3: The eleven comparison methods, organised by the question each answers about OARF. The column "Hedges over" names the uncertainty geometry; OARF is the only method whose set is an anisotropic *anchor-shaped* ellipsoid, and the only proactive (non-triggered) robust learner.

Method	Year	Family	Hedges over	Posture
OARF/OARF-CD	ours	anchor-robust OCO	anchor ellipsoid $\mathbb{E}[aa^\top]$	proactive
OGD	2003	basic baseline	—	—
Rolling-OLS	—	basic baseline	—	windowed
ACI	2021	conformal	coverage level	reactive
M-FISHER	2025	test-time adaptation	— (martingale trigger)	reactive
WATCH/WCTM	2025	conformal martingale	interval width	reactive
DR-Combo	2026	robust combination	Wasserstein ball	mirror descent
FC-DRO(-ES)	2026	robust combination	variance / ES	exponential weights
W-DRO-OL	2026	robust online learning	Wasserstein ball	primal–dual
CostACI	2026	cost-aware conformal	coverage level	reactive
AnchorReg / DRIG	'21/'23	causal (batch)	anchor / environments	windowed

How each method produces a forecast. To keep the comparison even, every method emits a one-step point forecast and is scored by the same squared loss under the same leak-free protocol. The linear learners (OARF, OARF-CD, OGD, Rolling-OLS, W-DRO-OL) and the combination methods (DR-Combo, FC-DRO/-ES) output $w_t^\top \phi(x_t)$ directly. The conformal methods are run with a quantile read-out: ACI and CostACI maintain 5/50/95% pinball heads and use the median head as the point forecast (their adaptive level governs only the interval), and WATCH/WCTM uses an OGD point head with a martingale-driven interval; only these three additionally report coverage, CRPS and width. M-FISHER uses its adapted point head. Thus all methods are comparable on point loss, and the interval-based ones are additionally compared on distributional metrics where defined.

Metrics, protocol and hyper-parameters. We report point accuracy (overall and root MSE; per-regime, worst-regime and post-changepoint MSE over 50-step windows); interventional robustness on the held-out grid (the curve MSE(ν) and the worst-case error); coverage, CRPS and interval width for the quantile heads; and, as a secondary check (Appendix), the economic value of the volatility forecasts via a vol-timing strategy [7]. Significance uses the pairwise Diebold–Mariano test [5] and the Model Confidence Set [9]. We use an expanding-window protocol: the first fifth of each series is a warm-up / hyper-parameter block and the remainder is evaluated; synthetic numbers average over *sixteen* seeds and real numbers over three rolling origins. The real panel ends on 22 June 2026; the 2026 portion is complete through that date and the evaluation window is the post-warm-up remainder. *Real-data regimes* are terciles of a past-only volatility-stress index s_t, the lagged 20-day mean of standardised log realised variance, partitioned at its in-sample 33/67%

quantiles into calm/normal/turbulent states; results are qualitatively unchanged under a two-state median split. *All hyper-parameters* are listed in Table 4 and were fixed a priori or selected once on the warm-up block; the held-out intervention grid is used only for evaluation, never for tuning ξ or any other parameter. No future information enters any forecast.

Table 4: Hyper-parameters. Values were fixed a priori or chosen once on the warm-up window; none were tuned on the evaluation period or the intervention grid.

Symbol	Meaning	Value
η	learning rate (AdaGrad base)	0.1
ξ	robustness penalty (headline OARF)	6 (synthetic), 1.5 (real)
β	EMA decay for moments	0.98
ϵ	ridge on the anchor Gram	10^{-3}
λ_2	ℓ_2 weight	10^{-4}
η_B	channel-discovery rate	0.3
L	discovery block length	40
γ_c	grand-mean penalty (discovery)	0.3
q	channel dimension	2 (synthetic), 3 (real)
winsorisation	standardised-feature clip	± 8
α	nominal miscoverage (conformal heads)	0.1
warm-up	fraction reserved for warm-up/HP	0.2

5 Results

Synthetic: fixed-channel robustness, and an unstable learned channel. In the controlled benchmark (Table 5, Figures 1–4), aggregated over 50 seeds and reported as medians with interquartile ranges, the in-distribution and interventional rankings are nearly reversed. *In distribution*, where every regime has been seen, the batch and reactive methods are most accurate: rolling-window OLS, DRIG and the conformal heads post the lowest median overall MSE (≈ 0.011), and fixed-channel OARF, which declines to use the anchor-correlated child signal, pays for that with a higher median in-distribution error (0.019). This is a real cost, not a rounding effect, and we do not minimise it. *Under intervention*, the order reverses: OARF attains a median worst-case interventional MSE of 0.041, about 57% below online gradient descent (0.095), 58% below the Wasserstein-DRO online learner (0.098) and WATCH/WCTM (0.099), and 80% below the reactive M-FISHER (0.204); it does not match but approaches the batch anchor-regression reference that is *given* the true anchor (0.026). The interquartile ranges overlap — the interventional metric is heavy-tailed and seed-dependent — so we read the gap as a robust shift in the median rather than a uniform per-seed dominance; the alignment-versus-robustness scatter (Figure 6) shows the relationship seed by seed. The robustness curve (Figure 3a) shows the mechanism — OARF and the batch reference stay nearly flat as the intervention magnitude grows while the reactive and agnostic-robust methods climb steeply — and sweeping ξ traces the trade-off (Figure 3b). The update directly targets the anchor–residual correlation, which OARF holds near zero while it spikes for OGD at every regime change (Figure 4).

The learned-channel variant is the weaker and less reliable part of the story, and we report it as such. OARF-CD recovers the true subspace on average (alignment 0.88 ± 0.16, Figure 5) and its *median* interventional robustness is competitive (0.049, close to fixed-channel OARF), but the distribution is *heavy-tailed*: the worst-case interventional MSE averages 0.147 across seeds, roughly three times its median, so on a non-trivial fraction of seeds the discovered channel is poor and the method approaches or exceeds doing nothing (OGD, median 0.095). The gap between its median and mean (Figure 6) is exactly the instability we want to flag: as an interventional-robustness method the learned channel is not yet a dependable substitute for a known one, and closing this tail is the main practical limitation of the discovery layer.

Table 5: Synthetic anchor-SCM over 50 seeds (lower is better). We report the median with the interquartile range $[q_{25}, q_{75}]$ in brackets for the overall and worst-do(A) columns, and the median for the worst-regime and post-changepoint columns; medians and IQRs are robust to the heavy right tail of the interventional metric. The final column is the held-out interventional metric; the best fully online method is in bold. AnchorReg/DRIG are *batch* methods, and AnchorReg is additionally given the true anchor, so they are references rather than competitors. OARF-CD's interventional median is competitive but its tail is heavy (mean 0.147), discussed in the text.

Method	overall MSE	worst-regime MSE	post-cp MSE	worst-do(A) MSE
OARF (fixed channel)	0.019 [0.010, 0.043]	0.030	0.021	**0.041 [0.015, 0.071]**
OARF-CD (learned B)	0.018 [0.010, 0.040]	0.029	0.021	0.049 [0.020, 0.124]
OGD	0.012 [0.008, 0.033]	0.022	0.018	0.095 [0.031, 0.220]
Rolling-OLS	0.010 [0.005, 0.030]	0.015	0.012	0.083 [0.025, 0.278]
ACI	0.011 [0.006, 0.030]	0.015	0.013	0.071 [0.022, 0.164]
M-FISHER (2025)	0.021 [0.016, 0.043]	0.043	0.039	0.204 [0.072, 0.387]
WATCH/WCTM (2025)	0.012 [0.008, 0.033]	0.022	0.018	0.099 [0.031, 0.220]
W-DRO-OL (Chen 2026)	0.012 [0.007, 0.033]	0.022	0.018	0.098 [0.033, 0.288]
CostACI (2026)	0.011 [0.006, 0.030]	0.015	0.013	0.071 [0.022, 0.164]
AnchorReg (batch, true anchor)	0.011 [0.005, 0.031]	0.014	0.011	0.026 [0.010, 0.074]
DRIG (batch)	0.010 [0.005, 0.030]	0.015	0.014	0.116 [0.032, 0.281]

The discovery layer: subspace recovery without stable robustness. Left to find its own anchor, OARF-CD recovers the true subspace *on average* — mean principal-subspace alignment 0.88±0.16 against a random-subspace baseline of 0.42, with individual seeds climbing toward 0.99 as the stream lengthens (Figure 5). But high mean alignment does not translate into stable robustness: the relationship between a seed's final alignment and its worst-do(A) MSE is noisy (Figure 6), and the seeds with transiently poor alignment dominate the heavy interventional tail reported in Table 5. The honest reading is that the layer succeeds at *nonstationary-subspace recovery* but not yet at delivering the fixed channel's reliable interventional robustness; it is a proof of concept for removing the analyst-supplied-anchor assumption, not a finished replacement for a known channel.

Ablations. Table 6 and Figure 7 take the method apart one component at a time, and the picture that emerges is that the anchor channel is doing the real work. With the true channel the worst-case interventional error is 0.080; replace it with a *random* channel and it climbs to 0.377 — worse than using no anchor penalty at all (OGD, 0.226). That a wrong channel is actively harmful, not merely useless, is evidence that identifying the right channel is the central issue, and it is the same phenomenon the stress tests quantify continuously below. The learned channel (0.303, alignment 0.84) sits near the OGD baseline on average and, as Table 5 shows, is unstable across seeds. The remaining components play supporting roles: AdaGrad matters for both accuracy and robustness (a tuned fixed step raises in-regime MSE from 0.035 to 0.095 and worst-do(A) from 0.080 to 0.145); EMA centring improves in-regime accuracy (0.035 vs. 0.043) at essentially unchanged robustness, consistent with separating the regime mean from the covariance the penalty uses. On

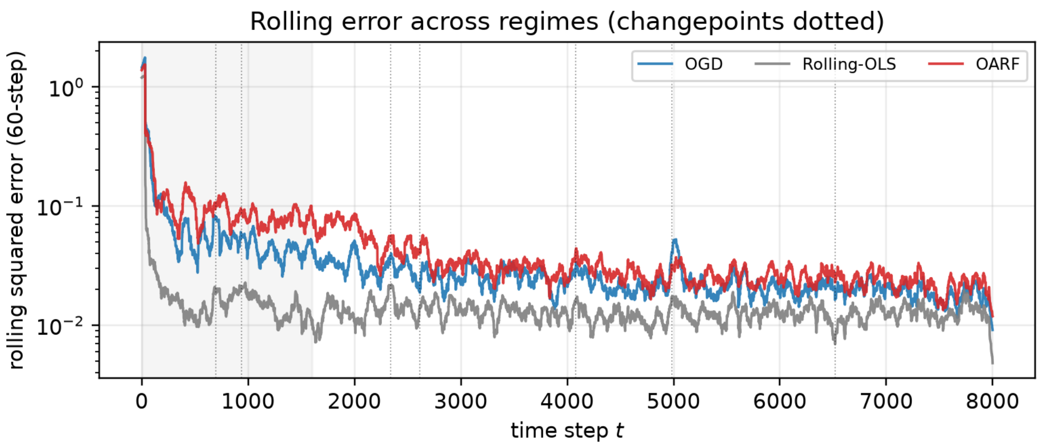


Figure 1: Rolling squared error across regimes on a representative seed (changepoints dotted, the shaded band is the warm-up window). The reactive baselines spike at every break; OARF stays comparatively flat.

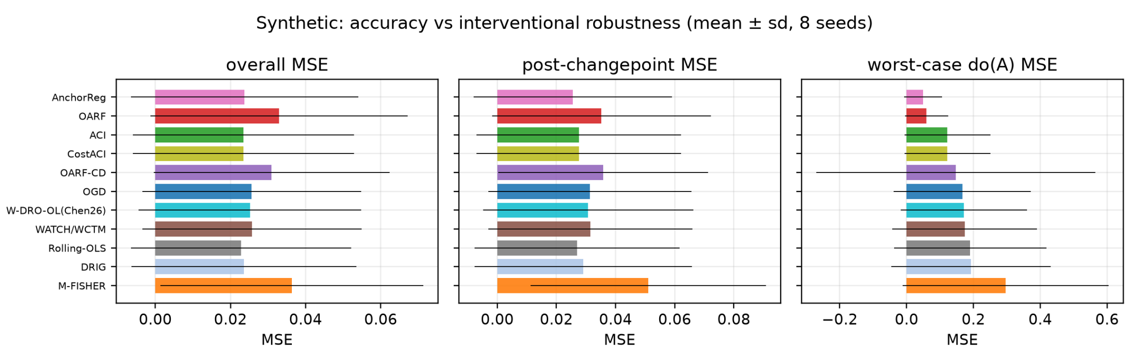
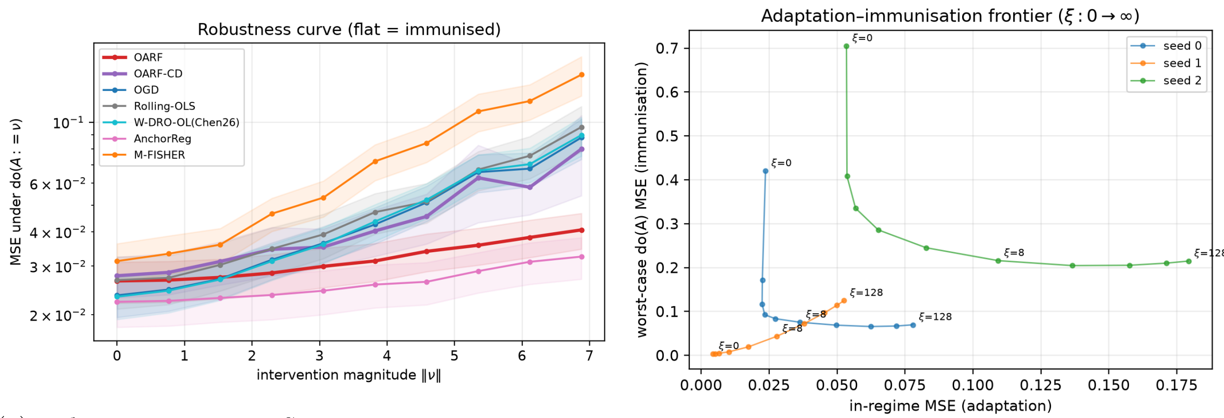


Figure 2: Overall, post-changepoint and worst-case interventional MSE on the synthetic benchmark, sorted by interventional robustness. The reactive and batch methods win in distribution; OARF and the batch oracle win under intervention — the two panels are nearly mirror images.



(a) Robustness curve: MSE against intervention magnitude $\|\nu\|$.

(b) Adaptation–robustness frontier as $\xi : 0 \to \infty$.

Figure 3: (a) OARF and the batch oracle remain near-flat as interventions grow, while reactive and agnostic-robust methods degrade steeply. (b) The penalty ξ dials the learner continuously from the online-gradient-descent corner to maximal robustness.

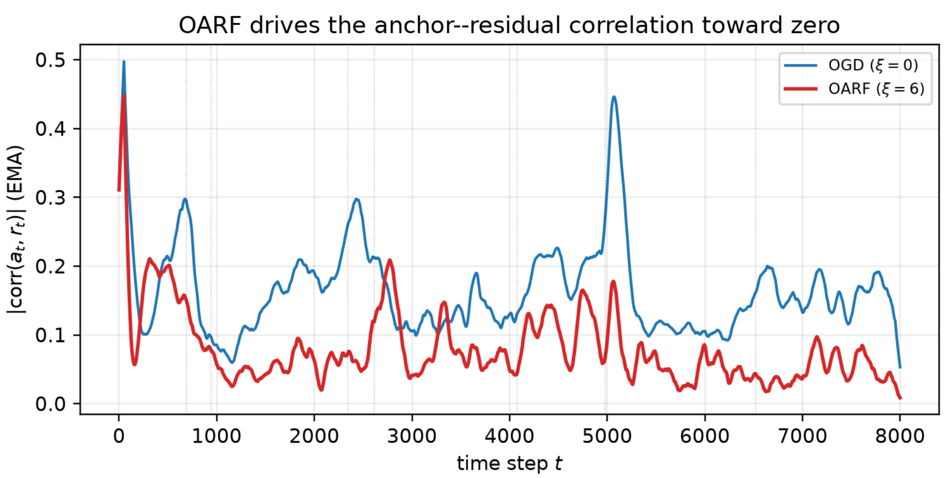


Figure 4: The mechanism, made visible. The smoothed anchor–residual correlation $|\mathrm{corr}(a_t, r_t)|$ is held near zero by OARF ($\xi = 6$) but spikes for online gradient descent ($\xi = 0$) at each regime change (dotted).

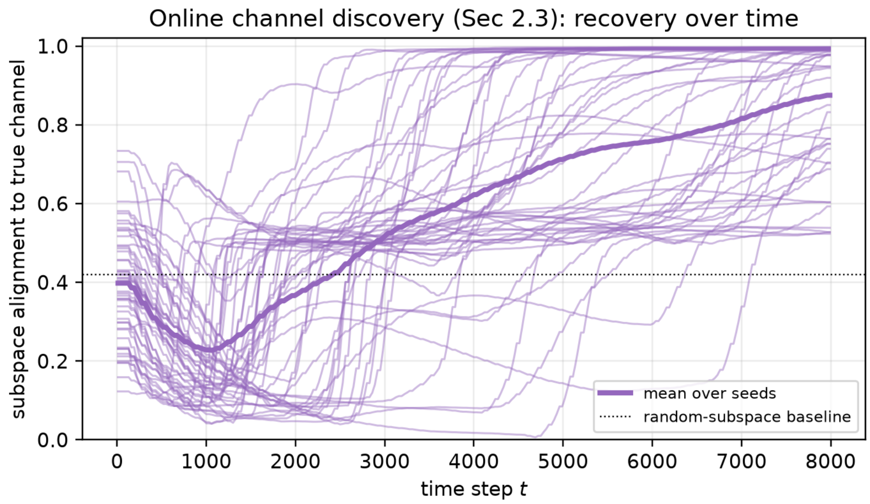


Figure 5: Online channel discovery: alignment of the learned channel B_t to the true intervention subspace over time. Thin lines are individual seeds, the bold line is their mean, and the dotted line is the random-subspace baseline.

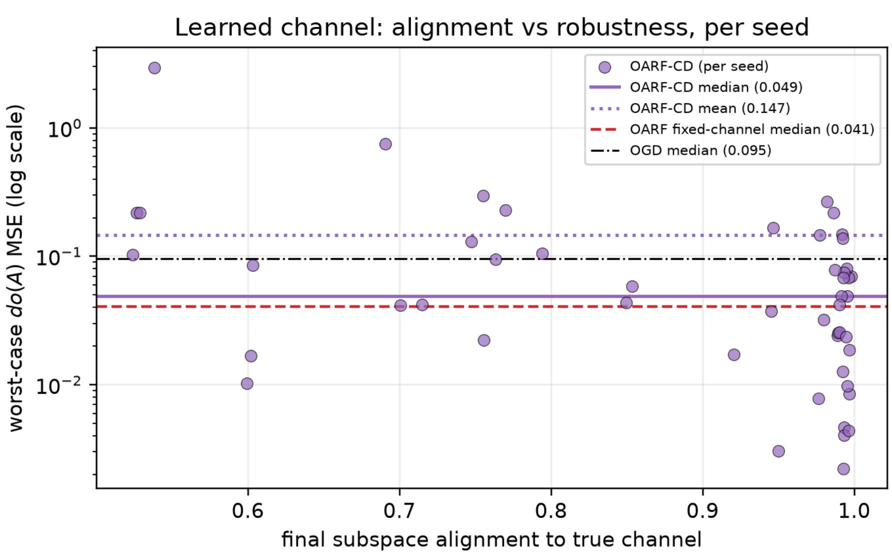


Figure 6: Per-seed alignment versus interventional robustness for the learned channel (50 seeds). Each point is one seed: the horizontal axis is the final subspace alignment of OARF-CD's discovered channel and the vertical axis its worst-case do(A) MSE (log scale). High-alignment seeds cluster at low error near the fixed-channel OARF median (dashed), but a minority of low-alignment seeds incur an order-of-magnitude higher error, pulling the mean (dotted) far above the median (solid) — the heavy tail that makes the learned channel unreliable in the worst case.

the discovered dimension, worst-do(A) MSE and alignment both improve gently as q rises from 1 to 4 ($0.329 \to 0.261$, alignment $0.79 \to 0.90$), because a larger learned subspace still contains the true two-dimensional channel; under-specifying ($q = 1$) is the more damaging error, as it cannot. The method is broadly insensitive to the EMA decay β over $[0.90, 0.995]$.

Table 6: Component ablations on the synthetic benchmark (mean over 16 seeds, $\xi = 6$ unless noted). The channel, not the optimisation machinery, carries the interventional robustness; a wrong channel is worse than none.

Variant	in-regime MSE	worst-do(A) MSE
OARF (full, true channel)	0.035	**0.080**
− EMA centring	0.043	0.081
− AdaGrad (fixed step)	0.095	0.145
random channel	0.032	0.377
$\xi = 0$ (OGD)	0.027	0.226
learned channel (OARF-CD, alignment 0.84)	0.033	0.303

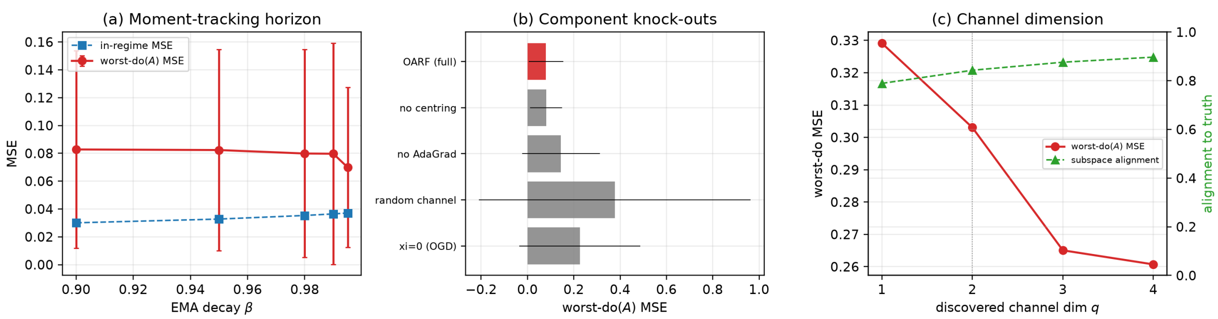


Figure 7: Ablations. (a) Sensitivity to the EMA decay β. (b) Component knock-outs by worst-case interventional MSE: the channel and the adaptive step are load-bearing, a random channel is worse than none. (c) Discovered-channel dimension: robustness and alignment both improve with q and are already strong at the true $q = 2$ (dotted).

Stress tests: when the anchor is wrong, OARF stops helping. The synthetic benchmark is, by construction, favourable: the anchor is clean and fully observed. The relevant question for practice is what happens when it is not, so we stress the fixed-channel method along two axes that break the ideal conditions (Figure 8; the synthetic SCM is otherwise unchanged, 8 seeds). *First*, we corrupt the observed context with additive noise of growing scale before any method sees it. Worst-do(A) MSE rises monotonically with the noise, and OARF crosses the no-anchor OGD baseline at roughly one standard deviation of observation noise (0.18 vs. OGD 0.19): beyond that point a noisy anchor is no better than no anchor, and the learned channel degrades sooner. *Second*, we rotate the supplied channel away from the truth by an angle θ (subspace alignment $\cos\theta$). OARF beats OGD while the alignment exceeds about 0.6 (roughly $\theta \leq 50°$) and is *worse* than OGD once the channel is more misaligned, reaching 0.36 versus OGD's 0.19 at an orthogonal channel — the same mechanism as the random-channel row of Table 6. The practical message is a usable diagnostic rather than a reassurance: OARF should be deployed only when the analyst has reason to believe the anchor is informative (alignment well above 0.6, noise below order one); when the channel is weak, noisy, or guessed, anchoring is not a free hedge and can actively hurt, and a noise-only or unstable-discovery diagnostic — e.g. monitoring whether the learned channel stabilises — should gate its use.

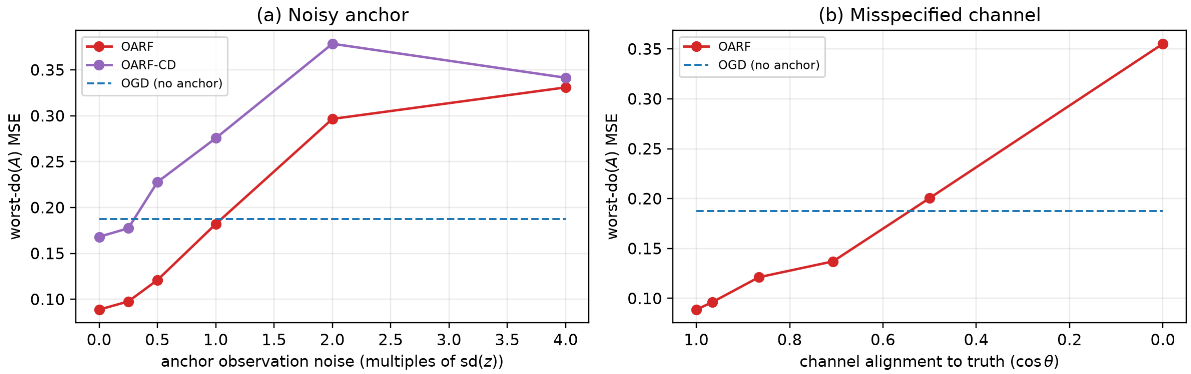


Figure 8: Stress tests of the fixed-channel method (worst-do(A) MSE; OGD baseline dashed). (a) Increasing anchor observation noise: OARF crosses the no-anchor baseline near one standard deviation of noise. (b) Rotating the channel away from the truth: anchoring helps only while alignment $\cos\theta \gtrsim 0.6$ and hurts beyond it. A wrong anchor is not a benign hedge.

Assumption-violation scenarios: where OARF helps and where it fails. The stress tests vary anchor *quality*; we also vary the *SCM structure* to break, one at a time, the assumptions behind Proposition 1 (Table 7, median worst-do(A) MSE over 8 seeds). OARF retains its advantage when the intervention also shifts the parents (PARENTS; the invariant predictor remains valid), when a second regime driver is *unobserved* (UNOBSERVED; immunising the observed part still helps), and when the observed anchor is a noisy *proxy* for the true latent driver (PROXY) — in each case it beats OGD, so the method degrades gracefully rather than catastrophically. But it genuinely *fails* in two cases: when the regime shifts the target's own parent-coefficient (MECHANISM), no fixed predictor is robust and both methods collapse ($\geq$1.0); and when regimes are *variance* shifts rather than mean shifts (VARIANCE), the mean-shift equivalence does not apply and OARF is *worse* than OGD. These are exactly the regimes the theory does not cover, and we report them as limits, not anomalies.

Table 7: Assumption-violation scenarios (median worst-do(A) MSE over 8 seeds; lower is better). OARF uses the true observed channel. It helps when it beats OGD and fails otherwise; it fails under a shifting target mechanism and under variance- rather than mean-shift regimes.

Scenario	OARF	OARF-CD	OGD	assumption broken
clean (ideal)	0.021	0.033	0.051	none (reference)
parents shifted	0.018	0.016	0.047	none (still anchor-mediated)
unobserved driver	0.024	0.040	0.048	channel incompletely observed
noisy proxy anchor	0.044	0.058	0.065	anchor predictive, not causal
mechanism shift	1.299	1.238	1.020	target mechanism non-invariant
variance shift	0.046	0.040	0.025	regimes shift variance, not mean

Is the discovery rule a good anchor selector? Because automatic anchor selection is the method's most fragile element, we ask whether OARF-CD's block-mean rule is a sensible choice by comparing it, on a strict pre-deployment budget (channel estimated on the *warm-up* block only, no look-ahead), against alternatives (Table 8): a supervised residual–context-coupling selector, stationary PCA of the context, and a random channel, all scored by alignment to the truth and by the resulting fixed-channel worst-do(A) MSE. The block-mean rule is competitive with the supervised selector and clearly better than stationary PCA, confirming it is reasonably motivated rather than arbitrary; but the more important finding is negative — from a short warm-up *no* unsupervised selector reliably separates from the random channel (0.10–0.12 worst-do(A) MSE vs. random 0.12, against the true channel's 0.07). High alignment is reached only by learning over the full stream (OARF-CD, alignment 0.84 in Table 5), and even then unstably. We conclude that reliable *online* anchor discovery from limited data is an open problem, and that practitioners should supply a candidate anchor and validate it (below) rather than rely on automatic discovery alone.

Table 8: Anchor selectors estimated on the warm-up block only (8 seeds; alignment to the true channel, and worst-do(A) MSE of the resulting fixed-channel OARF; lower MSE / higher alignment better).

Selector	subspace alignment	worst-do(A) MSE (median)
true channel (reference)	1.00	0.070
block-mean (OARF-CD rule)	0.42	0.101
supervised residual–context coupling	0.37	0.093
stationary PCA of context	0.31	0.236
random channel	0.43	0.115

Diagnostics and an anchor-selection procedure. Together the stress tests, scenarios and selector comparison yield a practical recipe rather than a black box. Before trusting OARF — and especially OARF-CD — a practitioner should: (1) propose candidate anchors from domain knowledge (for energy, the macro-uncertainty indices); (2) estimate the anchor–residual correlation of a plain online learner and check it is non-trivial and *stable* over the warm-up; (3) run a warm-up stress test, artificially perturbing the anchor as in Figure 8, and confirm that anchoring improves validation worst-regime loss without an unacceptable in-distribution cost; (4) for OARF-CD, monitor whether the learned subspace *stabilises* and aligns with interpretable drivers, and fall back to a fixed anchor or to $\xi = 0$ (plain OGD) if it does not. This turns the method's anchor-dependence from a hidden risk into an explicit, checkable precondition.

Real data: competitive, but not significantly better. On the energy panel (Table 9) the honest conclusion is that OARF is *competitive but not a clear winner*. The central fact is statistical: on both targets *every* listed method lies within the 10% Model Confidence Set (Diebold–Mariano statistics in Figure 10 are mostly small), so the differences in Table 9 are *not* statistically significant and should not be read as establishing superiority. Within that caveat the point estimates are mildly favourable to the anchor methods on the regime-robustness metrics — on Brent, OARF-CD has the second-lowest worst-regime MSE (0.229) behind the batch/conformal methods and below OGD (0.233), DR-Combo (0.233), FC-DRO (0.235) and M-FISHER (0.269); on Henry Hub OARF-CD has the lowest worst-regime error among the online learners (0.385) — but on *overall* MSE ACI, Rolling-OLS and AnchorReg are at least as good, and ACI is the strongest single method on several metrics. We therefore do not claim OARF "leads" here; we claim it is never dominated and is on par with the best baselines, while M-FISHER and the Expected-Shortfall combiner are the weakest. The most likely explanation is that real energy variance has weaker, noisier driver–residual coupling than the controlled SCM — precisely the regime in which the stress tests above predict the anchor advantage shrinks toward zero. The conformal head's coverage is near nominal (0.90). A decision-economic vol-timing check is deferred to the appendix: over this sample the risk-targeting Sharpe ratios are low and statistically indistinguishable across methods, so it carries no evidential weight.

Table 9: Real energy log daily-variance-proxy combination, 2010–2026 evaluation window (lower is better). Every listed method lies within the 10% Model Confidence Set; OARF and OARF-CD are competitive with the online robust baselines on worst-regime MSE; differences are within the MCS.

	Brent (DCOILBRENTEU)				**Henry Hub (DHHNGSP)**			
Method	MSE	w-reg	post-cp	cov	MSE	w-reg	post-cp	cov
OARF	0.2225	0.2313	0.2384	–	0.2870	0.3928	0.2670	–
OARF-CD	0.2225	0.2292	0.2433	–	0.2857	0.3850	0.2627	–
OGD	0.2243	0.2334	0.2412	–	0.2860	0.3888	0.2655	–
Rolling-OLS	0.2203	0.2230	0.2358	–	0.2882	0.3916	0.2623	–
ACI	0.2180	0.2221	0.2399	0.900	0.2808	0.3833	0.2426	0.904
M-FISHER (2025)	0.2680	0.2691	0.3307	–	0.3532	0.4428	0.3442	–
DR-Combo (Wang 2026)	0.2304	0.2325	0.2507	–	0.2891	0.3958	0.2432	–
FC-DRO (Liu 2026)	0.2318	0.2346	0.2571	–	0.2872	0.3916	0.2436	–
FC-DRO-ES (Liu 2026)	0.2387	0.2440	0.2713	–	0.3056	0.4086	0.2689	–
W-DRO-OL (Chen 2026)	0.2243	0.2334	0.2406	–	0.2859	0.3890	0.2633	–
AnchorReg	0.2183	0.2195	0.2345	–	0.2907	0.3942	0.2716	–

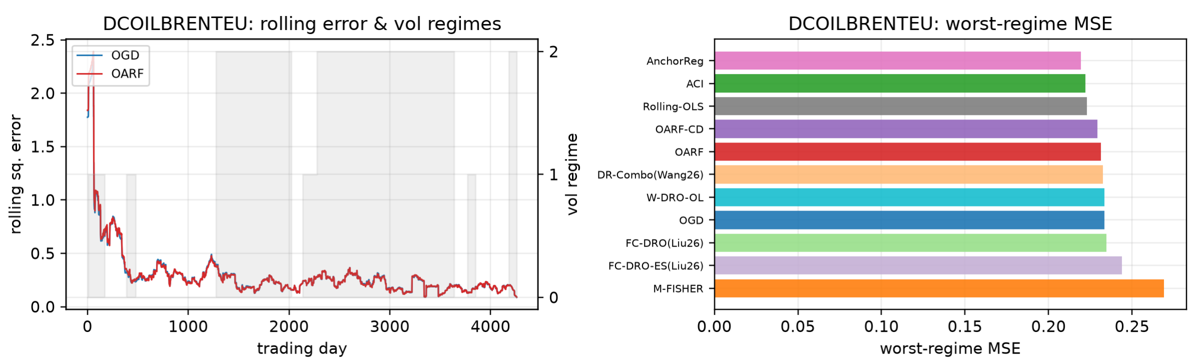


Figure 9: Brent: rolling squared error with the volatility-regime overlay (left) and worst-regime MSE by method (right).

Significance. The two Model Confidence Set analyses delineate what the evidence does and does not support. On the synthetic benchmark, the set computed on *in-distribution* loss (over 16 seeds) retains the batch and conformal methods — Rolling-OLS in all 16 seeds, DRIG in 15, the conformal heads in 11 — and excludes fixed-channel OARF (retained in 0 seeds) and OARF-CD (1 seed). This is the expected and correct outcome: OARF trades in-distribution accuracy for interventional robustness, which the in-distribution loss does not reward, so the in-distribution MCS is not the metric on which OARF should be judged — the held-out intervention grid is. On the real tar-

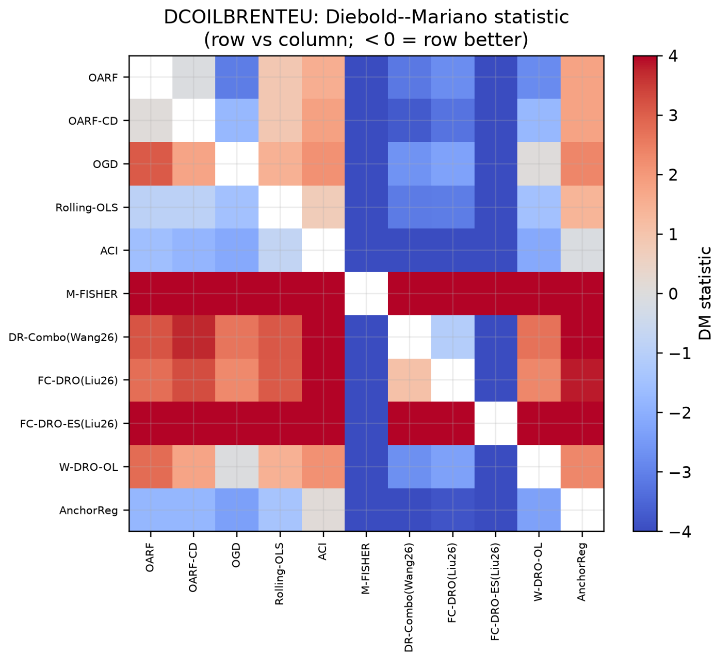


Figure 10: Brent: pairwise Diebold–Mariano statistics (negative = row more accurate than column). Most pairs are small in magnitude, consistent with the Model Confidence Set retaining all methods.

gets the MCS retains 10 of 11 methods on each commodity (only the weakest is excluded), so no method significantly out-predicts the field on average; we therefore treat the real-data differences as descriptive, not significant. The Diebold–Mariano statistics corroborate this — pronounced on the synthetic benchmark (Figure 11) and mostly small on the real targets (Figure 10).

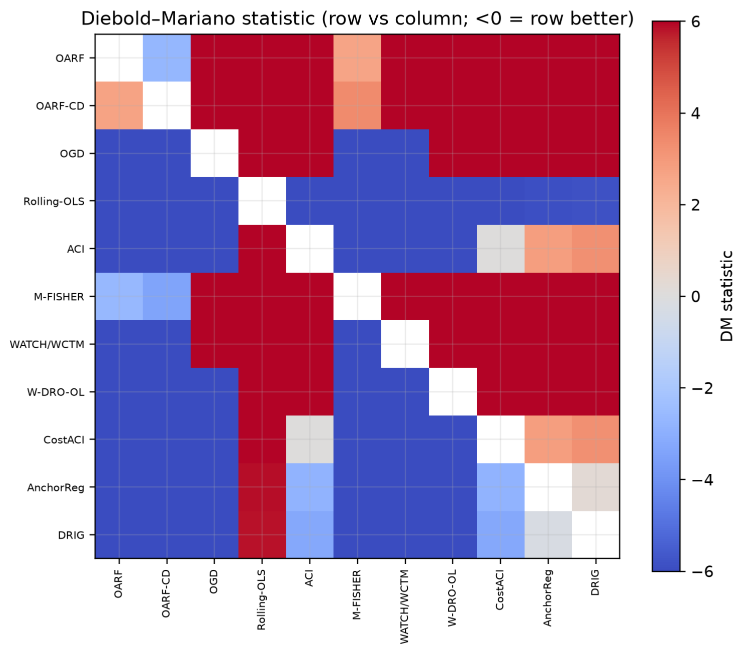


Figure 11: Pairwise Diebold–Mariano statistics on the synthetic benchmark; a negative entry means the row method is significantly more accurate than the column method.

6 Discussion and conclusion

The evidence supports a bounded claim. When regime shifts travel through an observable channel with anchor-mediated confounding, a forecaster can be made proactively robust to them advance and online, and the payoff is a large reduction in worst-case interventional error for a real but tunable increase in in-distribution error. When that structure is weak — as on the real energy series — the method is competitive but not significantly better than the strongest baselines. The geometric reading is that robustness is a budget: distributionally robust optimisation spends it isotropically, whereas anchoring concentrates it on the directions the regime actually moves; that concentration helps only to the extent that those directions are correctly identified.

When OARF helps and when it hurts. The same mechanism that helps under a good anchor hurts under a bad one, so we summarise the operating conditions explicitly.

OARF is likely to *help* when: the anchor is exogenous or approximately so; regime shifts are primarily mean shifts in the anchor; the anchor is observed with at most moderate noise; and the anchor–residual coupling is stable enough to estimate online (channel alignment $\gtrsim 0.6$).

OARF is likely to *hurt* (underperform plain OGD) when: the anchor is misspecified or weakly aligned; the dominant regime driver is unobserved; shifts are not anchor-mediated (e.g. the target mechanism or the anchor *variance* shifts); the learned channel is unstable; or the stream is too short / regimes change faster than the EMA window can track.

Safeguard: monitor channel stability and the anchor–residual coupling online and fall back to $\xi = 0$ (plain OGD) when they indicate a weak or shifting channel.

Conclusion. We close with the lessons rather than a restatement. *What we learned:* fixed-channel online anchor regression can deliver genuine proactive robustness to regime shifts — a fixed-anchor interventional-regret guarantee and a large reduction in worst-case interventional error — but only when the anchor is exogenous-like, informative, and observed with moderate noise, and only for mean-shift regimes; outside that envelope it offers no benefit and can do harm. *What we did not solve:* reliable *online* discovery of the anchor channel — the block-mean rule is reasonable but unstable, and from short data no unsupervised selector beats a random channel, so automatic anchor discovery remains open. *Practical recommendation:* treat OARF as a tool to be used *with* diagnostics — propose a candidate anchor, validate it on a warm-up stress test, and fall back to plain online learning when the anchor proves weak or unstable — rather than a drop-in replacement. *Future work:* stabilising channel discovery, nonlinear anchors and predictors, richer (variance and mechanism) interventions, commodity-specific implied-volatility data, and event-triggered deployment monitoring. Methodologically, the lasting contributions are a streaming anchor-regression update with a fixed-channel guarantee, interventional regret as an evaluation lens, and a careful empirical account of when anchor-based proactive robustness helps or fails.

Open theoretical question. The fixed-channel theorem's EMA-moment rate of $T^{2/3}$ rests, through Lemma 3, on within-regime stationarity (Assumption 4); tightening it toward $\sqrt{T}$ without a changepoint trigger, and extending any guarantee to the learned channel, are the main open problems.

Reproducibility. All code and data-build scripts that regenerate every number, table and figure in this paper are available at `https://github.com/vinhqdang/OARF-Online-Anchor-Robust-Forecasting-under-Economic-Regime-Shift`.

A Proof of Theorem 1

Throughout, $\mathcal{R}_t(w) := \sup_{\|\nu\|\le\rho(\xi)} \mathbb{E}_{\mathrm{do}(a:=\nu)}[(y_t - w^\top\phi(x_t))^2]$ is the per-step worst-case risk in (6), and $\hat{g}_t$ is the (inexact) step direction of Algorithm 1.

Lemma 1 (Convex surrogate and gradient decomposition)**.** *$\mathcal{R}_t$ is convex, and under Assumption 2 it equals, within the active regime, the anchor-penalised risk $\mathcal{R}_t(w) = \mathbb{E}_t[(y - w^\top\phi)^2] + \xi P_t(w)$ of (1). Moreover $\hat{g}_t = \nabla\mathcal{R}_t(w_t) + \zeta_t$ with $\zeta_t = M_t + b_t$, where $M_t := -r_t\phi(x_t) - \mathbb{E}_t[-r\phi]$ is a martingale difference ($\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] = 0$) and b_t is the deterministic ($\mathcal{F}_{t-1}$-measurable) error of the EMA-estimated penalty gradient relative to $\nabla(\xi P_t)$.*

Proof. For fixed ν, $\mathbb{E}_{\mathrm{do}(a:=\nu)}[(y - w^\top\phi)^2]$ is a convex quadratic in w; a pointwise supremum of convex functions is convex. The identity is Proposition 1 within the stationary active regime. The prediction term $-r_t\phi(x_t)$ is an unbiased one-sample estimate of $\frac{1}{2}\nabla\mathbb{E}_t[(y - w^\top\phi)^2] = -\mathbb{E}_t[r\phi]$, with centred part M_t; the robustness term substitutes the $\mathcal{F}_{t-1}$-measurable EMA moments for the population moments in $\nabla(\xi P_t)$, and b_t is that substitution error. □

Lemma 2 (Inexact dynamic regret)**.** *Run projected OGD with the path-length-adaptive Ader wrapper [20] on the convex losses $\mathcal{R}_t$, fed $\hat{g}_t$. Then for any $u_{1:T} \subset \mathcal{W}$ of path length P_T, $\sum_t \mathcal{R}_t(w_t) - \sum_t \mathcal{R}_t(u_t) \le O(\sqrt{T(1+P_T)}) + \sum_t\langle\zeta_t, w_t - u_t\rangle$, with the first term attained without prior knowledge of P_T.*

Proof. With exact gradients, Ader attains minimax dynamic regret $O(\sqrt{T(1+P_T)})$ for convex G-Lipschitz losses over a radius-D set, adaptively in P_T [20], subsuming Hazan [10], Zinkevich [21]. Convexity gives $\mathcal{R}_t(w_t) - \mathcal{R}_t(u_t) \le \langle\nabla\mathcal{R}_t(w_t), w_t - u_t\rangle = \langle\hat{g}_t, w_t - u_t\rangle - \langle\zeta_t, w_t - u_t\rangle$; summing and applying the exact-gradient bound to the $\hat{g}_t$ terms yields the inexactness term. □

Lemma 3 (Gradient-error control). *Under Assumptions 1–4, against the deterministic minimising comparator: (i)* $\mathbb{E}[\sum_t \langle M_t, w_t - u_t \rangle] = 0$*; and (ii)* $\mathbb{E}\|b_t\| \le c_1\sqrt{1-\beta}$ *at steps at least* $1/(1-\beta)$ *into a regime, with* $\|b_t\| \le c_2$ *within* $1/(1-\beta)$ *steps of each of the* K *changes, so* $\sum_t \mathbb{E}\|b_t\| \le c_1 T\sqrt{1-\beta} + c_2 K/(1-\beta)$*; with oracle moments* $b_t \equiv 0$.

Proof. (i) w_t is $\mathcal{F}_{t-1}$-measurable and, the schedule being oblivious (Assumption 3), u_t is deterministic, so $\mathbb{E}[\langle M_t, w_t - u_t \rangle \mid \mathcal{F}_{t-1}] = 0$; the tower rule concludes. (ii) An EMA of decay β of a stationary sequence has variance $\Theta(1-\beta)$ about its mean and geometrically-decaying within-regime bias; by Assumption 1 the penalty gradient is Lipschitz in these moments (the ϵ-ridge bounds the inverse), inheriting the $\Theta(\sqrt{1-\beta})$ steady-state scale. After a change the EMA mixes the previous regime for $O(1/(1-\beta))$ steps with bounded $\|b_t\| \le c_2$. Summing gives the bound; oracle moments make the substitution error vanish. □

Proof of Theorem 1. Combining the lemmas and taking expectations, for the deterministic minimiser $u_{1:T} \in \mathcal{W}$ ($P_T \le 2DK$), $\mathbb{E}[\sum_t \mathcal{R}_t(w_t)] - \sum_t \mathcal{R}_t(u_t) \le O(\sqrt{T(1+K)}) + 0 + D\sum_t \mathbb{E}\|b_t\|$. With oracle moments the last term vanishes, giving (i); with EMA moments it is $\le D(c_1 T\sqrt{1-\beta} + c_2 K/(1-\beta))$, which at $1-\beta_T = T^{-2/3}$ is $O((1+K)T^{2/3})$ and dominates $\sqrt{T(1+K)}$, giving (ii). Finally, by Assumption 2 the realised regime is an admissible intervention, so $\mathbb{E}_t[\ell_t(w_t) \mid \mathcal{F}_{t-1}] \le \mathcal{R}_t(w_t)$ and hence $\mathbb{E}[\sum_t \ell_t(w_t)] \le \mathbb{E}[\sum_t \mathcal{R}_t(w_t)]$; as $\min_{u\in\mathcal{W}} \sum_t \mathcal{R}_t(u)$ is the comparator in (6), the displays bound $\mathbb{E}[\mathrm{IntReg}_T(\rho(\xi))]$ by the stated rates. □

B Decision-economic evaluation (secondary)

As a secondary check we convert each method's volatility forecast into a volatility-timing portfolio [7]: exposure is scaled inversely to the forecast variance, $w_t = \mathrm{clip}(\sigma^{\star 2}/\hat{\sigma}_t^2, 0, 3)$, and the strategy earns $w_t r_{t+1}$ net of a one-basis-point turnover cost; we report the annualised Sharpe ratio, maximum drawdown and turnover. Over 2010–2026 these commodity risk-targeting Sharpe ratios are low and statistically indistinguishable across methods (Figure 12), so this evaluation carries no evidential weight and is reported only for completeness; it is not used to support any claim in the main text.

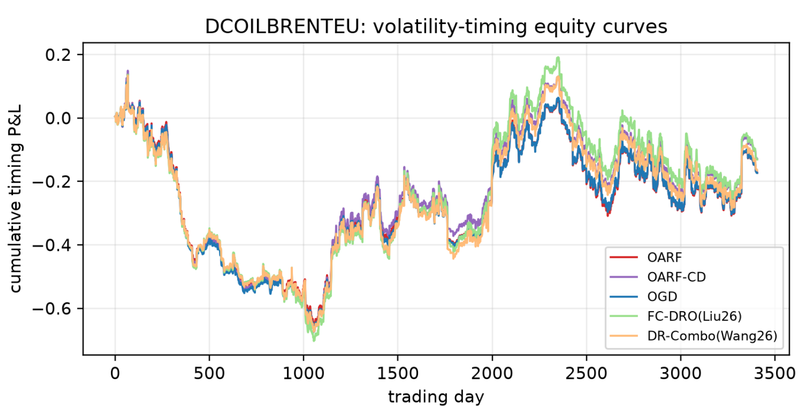


Figure 12: Brent: cumulative volatility-timing profit and loss. Differences across methods are economically small and not significant over this sample.

C Baseline implementations and notes

Table 10 records, for every method, the loss optimised, the update rule, what it outputs, and whether it is online/rolling/batch and uses the anchor; all run under the identical leak-free protocol with the hyper-parameters of Table 4. Per step the linear methods cost $O(d)$; OARF adds $O(pq + q^3)$ for the anchor block; Rolling-OLS and the batch methods cost $O(d^3)$ at each refit (every 5–10 steps). A few implementation points. The warm-up block is used *only* for past-only standardiser initialisation and one-off hyper-parameter selection, never for separate batch pre-training. Missing FRED observations and exchange holidays are handled by the business-day reindex with a five-day capped forward-fill; longer gaps are left as missing and skipped. The GARCH$(1, 1)$ expert uses fixed RiskMetrics-style coefficients after the warm-up rather than online re-estimation, so it is a fixed-parameter recursion. We use a larger channel dimension on the real data ($q = 3$ vs. $q = 2$ synthetic) because three macro-uncertainty drivers (VIX, GPR, EPU) are designated as the hand-chosen anchor, whereas the synthetic SCM has exactly two anchor directions by construction. OARF is presented for linear predictors; it extends to nonlinear forecasters by applying the same residual–anchor penalty to a feature map or to the top layer of a network, which we leave to future work.

Table 10: Baseline implementations. "Out" = P(oint)/I(nterval); "Mode" = O(nline)/R(olling)/B(atch); "Anc" = uses the anchor.

Method	Loss optimised	Update rule	Out	Mode	Anc
OARF/OARF-CD	anchor-penalised sq. err.	AdaGrad on (4)	P	O	yes
OGD	squared error	AdaGrad ($\xi = 0$)	P	O	no
Rolling-OLS	windowed ridge	closed-form refit	P	R	no
ACI/CostACI	pinball (3 quantiles)	subgrad. + ACI level	P+I	O	no
M-FISHER	sq. err. + martingale	OGD + triggered NGD	P	O	no
WATCH/WCTM	sq. err. + WCTM	OGD + width adapt	P+I	O	no
DR-Combo	Wasserstein-DR combo	EG mirror descent	P	O	no
FC-DRO(-ES)	variance-/ES-scaled	exponential weights	P	O	no
W-DRO-OL	Wasserstein-DR sq. err.	primal–dual OGD	P	O	no
AnchorReg	anchor-penalised (batch)	windowed closed-form	P	R	yes
DRIG	invariant-gradient (batch)	windowed two-env. fit	P	R	(env)

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